When a KPI Lies

A Data Analyst Case Study on Denominator Sensitivity in U.S. Immigration Admissions Data

Kai A. Sanchez P.

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Executive Summary

This project audits a percentage-based KPI built from administrative immigration data: the recorded blue-collar share of immigrant admissions to the United States.

Using a reproducible country-year panel for 19 Latin American origin countries from 1982 to 2000, the analysis asks whether higher IRCA (Immigration Reform and Control Act) legalization exposure is associated with a meaningful decline in the measured blue-collar share during the 1989–1992 surge years.

The main result is simple and important:

1. The estimated blue-collar decline is large when the denominator uses known occupations only.
2. The same result becomes much smaller when the denominator is expanded to include records with no occupation.
3. The best interpretation is therefore denominator instability and reporting sensitivity, not clear evidence of a true collapse in blue-collar admissions.

This makes the project a data analyst case study in KPI definition, denominator auditing, missingness, and responsible interpretation of administrative data.

Why This Works as a Data Analyst Project

This is not just a historical policy project, but also a measurement project. A reporting team could easily track a metric like "blue-collar share," see it fall during a major policy surge, and conclude that immigrant composition changed sharply. This project shows why that can be misleading when:

- denominator definitions are unstable,
- missingness changes over time,
- administrative coverage shifts during operational surges.

The broader lesson is:

> When the denominator is unstable, a percentage KPI can tell the wrong story.

That is why this project is best framed as a case study in:

- KPI design,
- denominator sensitivity,
- administrative data quality,
- and robust interpretation under imperfect measurement.

Business Problem

Suppose an analyst is asked:

Did immigrant occupational composition change sharply during the IRCA surge?

A naive answer would focus only on the net blue-collar share. A better analyst would asks:

- What exactly is in the denominator?
- Did coverage change at the same time?
- Is the share moving because the numerator changed, or because the denominator changed faster?
- Are "missing" or "no occupation" records affecting the interpretation?

That is the problem this project solves.

Data and Exposure Design

The project builds a harmonized country-year panel from public INS (Immigration and Naturalization Service) Statistical Yearbooks. It merges four inputs:

- occupation
- class of admission
- gender/age
- adjustments versus new arrivals

The final sample covers 19 Latin American origin countries from 1982 to 2000.

Exposure measure

Country-level IRCA exposure is defined as:

$$\text{Exposure} = \frac{\text{IRCA legalizations (1989–1992)}}{\text{Pre-IRCA admissions (1982–1985)}}.$$

Countries with larger legalization surges relative to their earlier admissions base are treated as more highly exposed.

Modeling approach

The core specifications are descriptive two-way fixed effects panel regressions with:

- country fixed effects,
- year fixed effects,
- clustered standard errors by origin country.

These estimates are interpreted as descriptive exposure-gradient associations, not causal treatment effects.

The Main Insight

The most important test in the project is a same-sample comparison between two versions of the same KPI:

- **Net denominator:** blue-collar count divided by known occupations only
- **Full denominator:** blue-collar count divided by known occupations plus no-occupation records

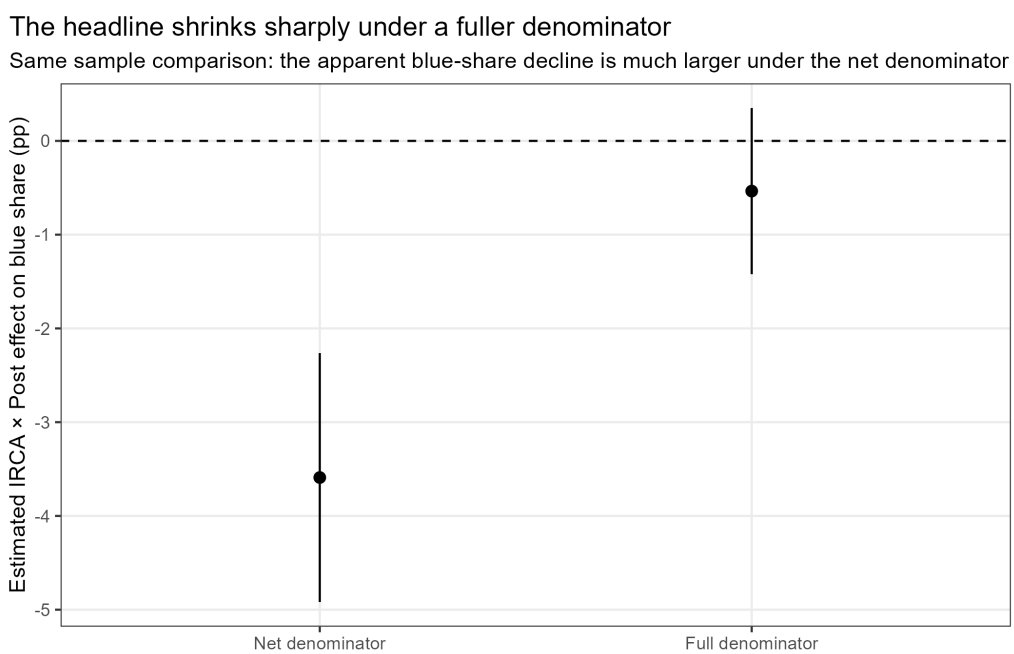

Figure 1: Main comparison: the estimated blue-share decline is much larger under the net denominator than under the full denominator.

Table 1: Main result: blue-collar share using the net versus full denominator on the same sample.

	Blue share (Net denom; same sample)	Blue share (Full denom; same sample)
IRCA intensity × Post	−0.036***	−0.005
	(0.007)	(0.005)
Adjustment share	0.169*	0.127**
	(0.085)	(0.059)
Num.Obs.	351	351

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by origin-country. Same-sample comparison: rows require both blue_share and blue_fullden observed. Portfolio takeaway: denominator sensitivity suggests reporting/missingness mechanics are central.

This is the core finding. The headline decline is substantial under the net denominator, but it shrinks sharply under the fuller denominator. Because both specifications use the same sample, the contrast is not driven by changing rows. It is driven by the definition of the KPI itself.

That makes the analytical takeaway clear: the apparent drop in blue-collar share is highly sensitive to denominator construction.

Magnitudes

The project also reports standardized effect sizes under both denominator definitions.

Table 2: Standardized effect sizes for blue-collar share using the net denominator.

Magnitude	Effect (pp / %)
Per 1 unit exposure	-3.59
Per 1 SD exposure	-7.61
Per IQR exposure	-2.46
As % of mean(Y): 1 SD	-22.22
As % of mean(Y): IQR	-7.18

Table 3: Standardized effect sizes for blue-collar share using the full denominator.

Magnitude	Effect (pp / %)
Per 1 unit exposure	-0.54
Per 1 SD exposure	-1.13
Per IQR exposure	-0.37
As % of mean(Y): 1 SD	-8.63
As % of mean(Y): IQR	-2.79

These magnitudes reinforce the same message as the main comparison: the effect is much larger when the KPI uses the narrower denominator.

Dashboard View of the Story

A good dashboard should not show only one headline KPI. It should show the surrounding context that helps interpret it correctly.

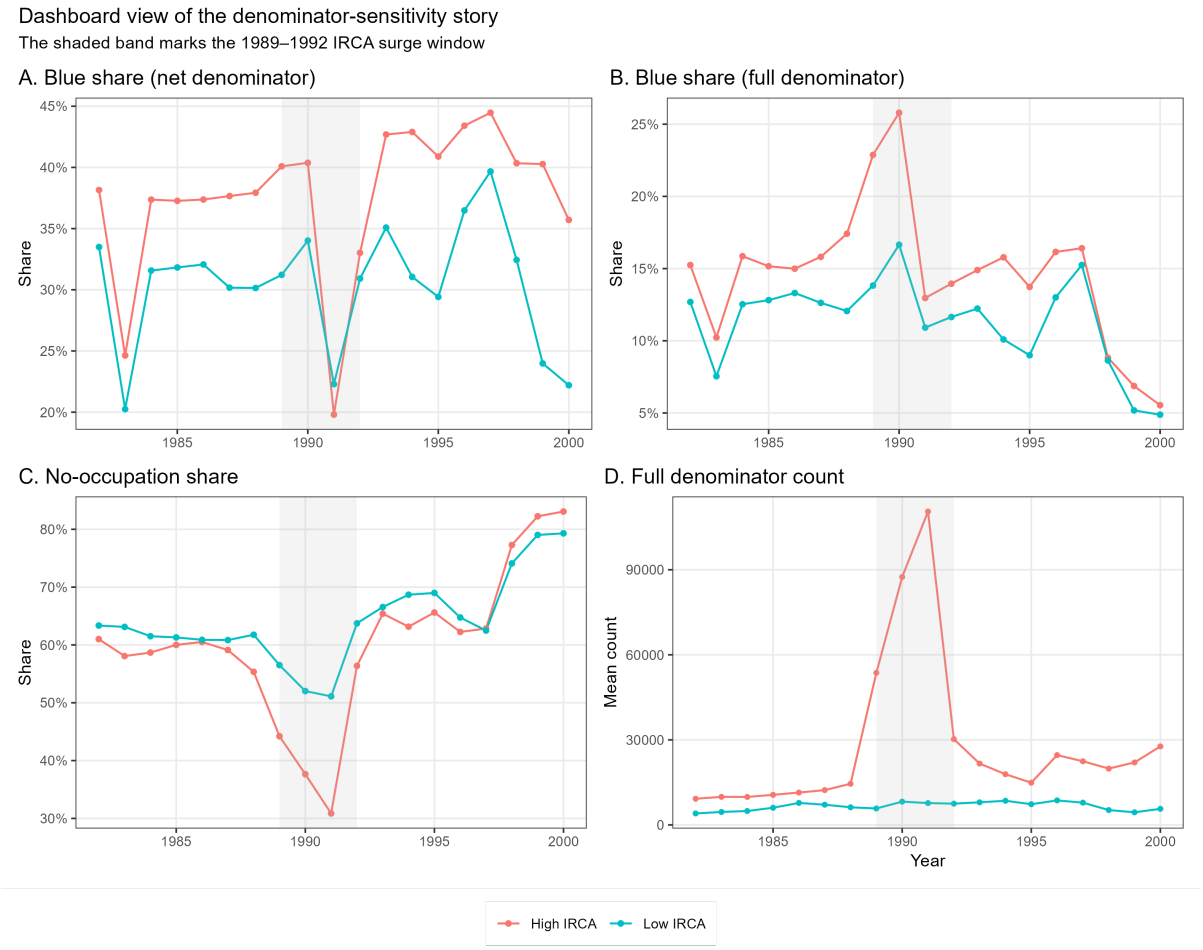

Figure 2: Dashboard view of the denominator-sensitivity story. The shaded region marks the 1989–1992 IRCA window.

Taken together, the dashboard shows why the single headline share can be misleading. The net share, full share, no-occupation share, and denominator count do not tell the same story, and that divergence is precisely the point of the project.

Looking Under the Hood: Numerator vs Denominator

A falling share can happen for at least two very different reasons:

- the numerator falls,
- or the denominator rises faster than the numerator.

To separate those possibilities, the project decomposes the share into underlying count components.

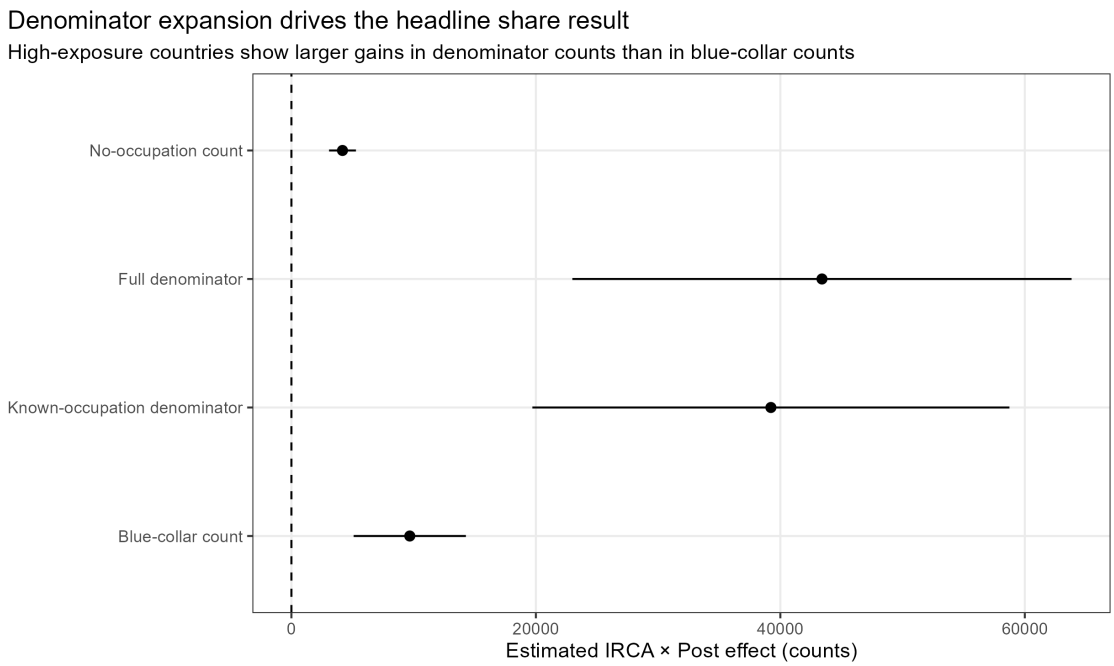

Figure 3: Decomposition in levels: denominator counts rise more strongly than the blue-collar numerator.

Table 4: Decomposition in levels.

	Blue level	Occ known level	Occ total level	No-occ level
IRCA × Post	9682.575***	39 226.652***	43 404.916***	4178.264***
	(2342.817)	(9956.342)	(10 418.047)	(561.725)
Adjustment share	−973.413	−18 202.159	−204.179	17 997.980
	(5699.283)	(21 227.495)	(32 984.662)	(17 559.005)
Num.Obs.	352	352	352	352

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by country. Levels are counts.

The decomposition is one of the strongest parts of the project. The blue-collar numerator does not simply collapse. Instead, denominator counts expand strongly in higher-exposure settings, which makes the share fall mechanically.

That is the move that turns this from a regression project into a real analytics audit.

Proportional version

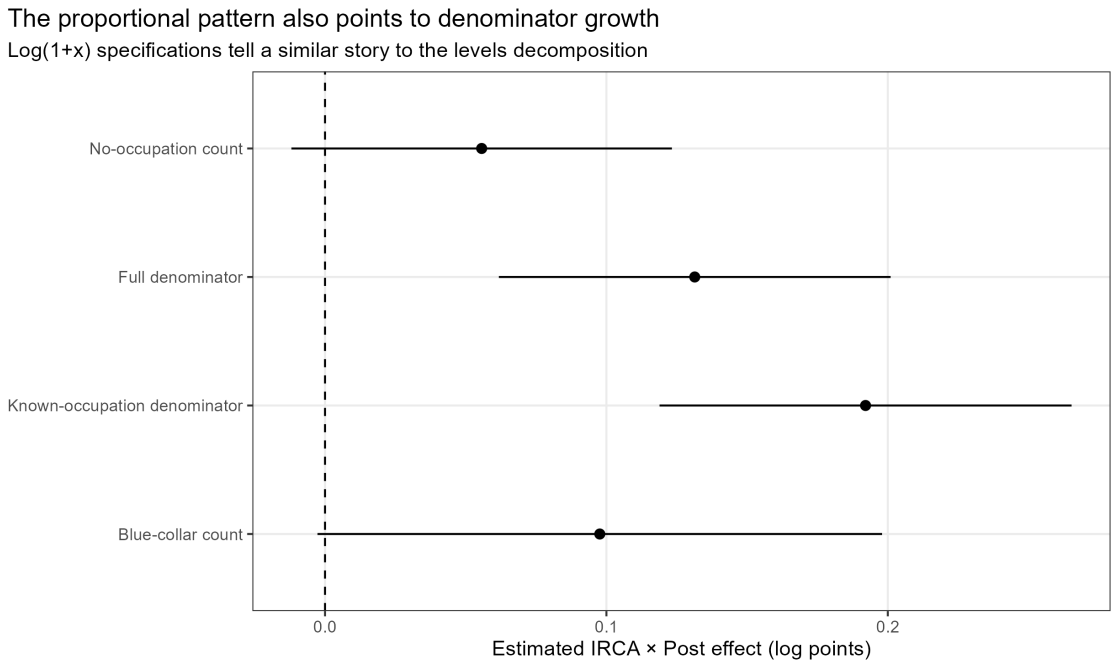

Figure 4: Decomposition in log(1+x): the proportional pattern also points to denominator growth.

Table 5: Decomposition in log(1+x).

	ln(1+Blue)	ln(1+Occ known)	ln(1+Occ total)	ln(1+No-occ)
IRCA × Post	0.098*	0.192***	0.131***	0.056
	(0.051)	(0.037)	(0.036)	(0.034)
Adjustment share	1.092	0.776	0.608	0.224
	(0.636)	(0.469)	(0.470)	(0.513)
Num.Obs.	352	352	352	352

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by country. Log specs use ln(1+x).

The proportional specifications tell a similar story, which increases confidence that the main interpretation is not an artifact of scale alone.

Denominator Channel

A natural denominator diagnostic is the behavior of the no-occupation category.

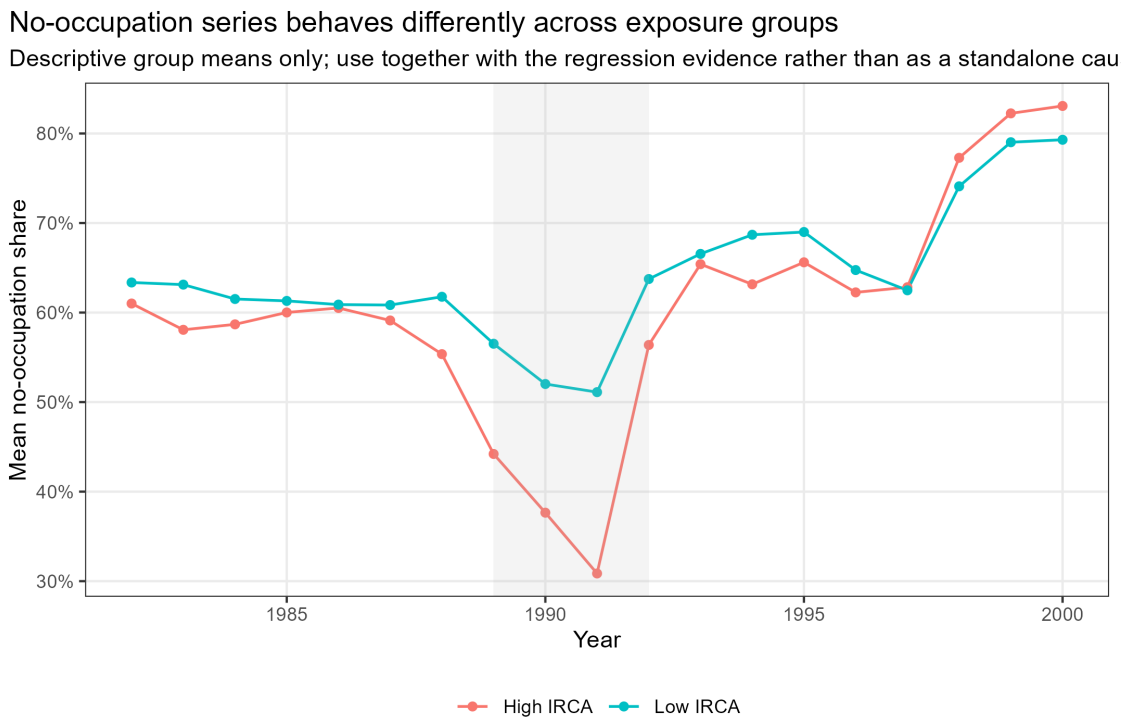

Figure 5: No-occupation share over time by IRCA exposure group. This is descriptive evidence and should be interpreted jointly with the regression results.

Table 6: Regression evidence for the no-occupation share on the net sample.

	No-occ share (Net sample)
IRCA × Post	−0.025***
	(0.005)
Adjustment share	−0.149*
	(0.081)
Num.Obs.	351

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by country.
Outcome: noocc_share.

This is an important nuance in the final version of the project: the descriptive trend figure and the regression evidence should be read together, not as a simple monotonic story. The safe interpretation is that denominator composition is unstable around the surge years, which is exactly why the blue-share KPI must not be interpreted in isolation.

Robustness

The project does not need endless robustness tables, but it does need enough evidence to show that the main pattern is not a fragile one-off.

Core robustness comparison

Table 7: Comparison of key robustness specifications for the blue-collar share.

	Baseline Net	Country trends	WLS admissions	Balanced Net
IRCA × Post	−0.036***	−0.036***	−0.036***	−0.037***
	(0.007)	(0.008)	(0.003)	(0.007)
Adjustment share	0.169*	0.160	0.150**	0.190*
	(0.085)	(0.099)	(0.069)	(0.090)
Num.Obs.	351	351	351	323

* p <0.1, ** p <0.05, *** p <0.01
Comparison of key robustness specifications for blue-collar share.

The result remains visible across key specifications such as country trends, weighting, and balanced-panel restrictions.

Additional sensitivity checks

Table 8: Additional sensitivity checks for the blue-collar share.

	Baseline Net	Drop 1995–1997	Drop UY+PY	Window 1982–1995
IRCA × Post	−0.036***	−0.042***	−0.037***	−0.037***
	(0.007)	(0.008)	(0.007)	(0.008)
Adjustment share	0.169*	0.224**	0.190*	0.182*
	(0.085)	(0.092)	(0.090)	(0.092)
Num.Obs.	351	300	323	260

* p <0.1, ** p <0.05, *** p <0.01
All models: Country & Year FE; SE clustered by country. Outcome: blue_share.

These checks matter because they show the signal is not entirely dependent on one exact coding choice or time window.

Leave-one-country-out

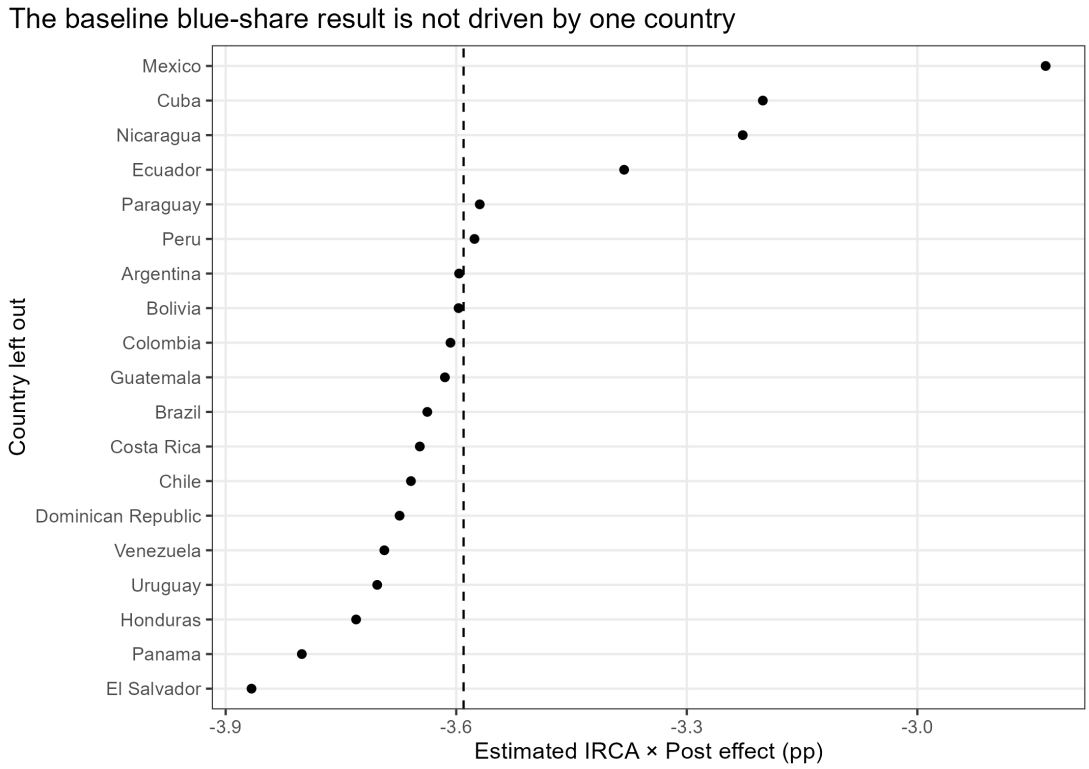

Figure 6: Leave-one-country-out influence on the baseline blue-share estimate.

The leave-one-country-out exercise shows that the baseline result is not driven by one single country.

Small-cluster inference

Table 9: Wild cluster bootstrap inference for the baseline net-denominator specification.

Model	Parameter	Bootstrap p-value	Bootstrap t-stat
lm with country+year FE (dummies)	IRCA intensity × Post	0.03361	-5.154

Because the panel is small at the country-cluster level, the wild bootstrap is a useful inference check.

Timing and Interpretation Limits

This project is descriptive, not causal. That is an important strength, not a weakness, because the report makes the limitation explicit rather than hiding it.

Event-study timing check

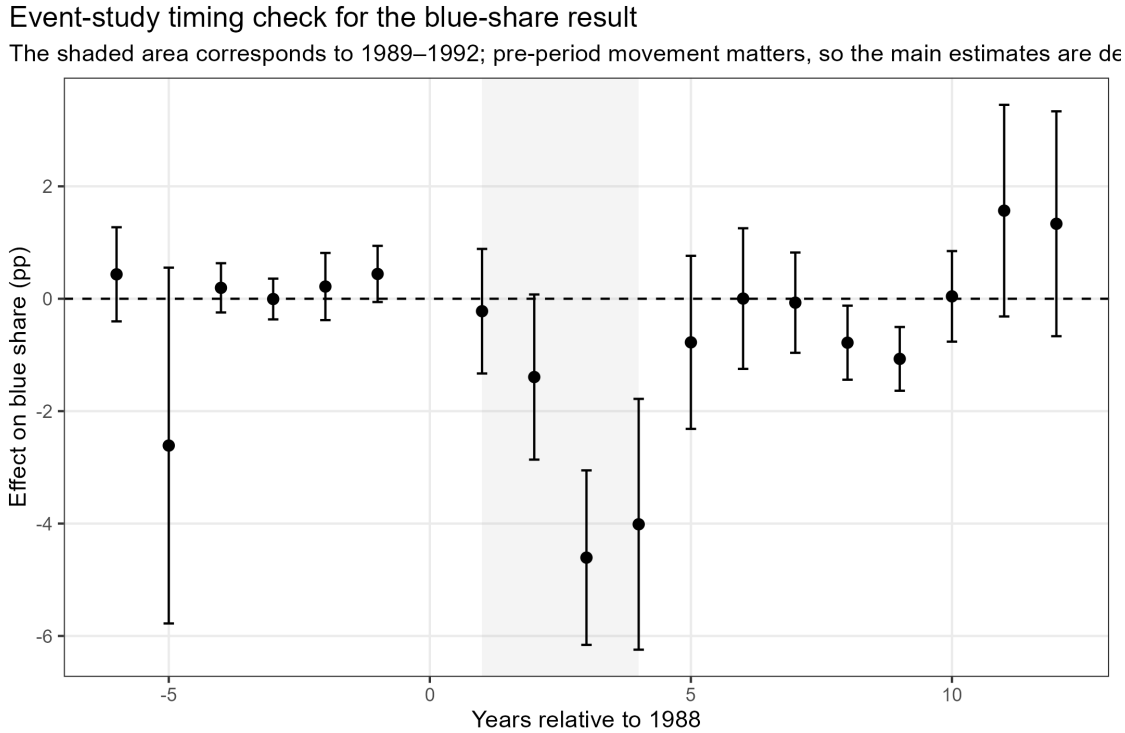

Figure 7: Event-study timing check for the blue-share result. The shaded area corresponds to 1989–1992.

Table 10: Wald test of joint pre-trend coefficients in the event-study specification.

Test	Statistic	df1	df2	p_value
Wald joint pre-trends (years < 1988)	13.54	6	298	0

The timing diagnostics show that higher-exposure countries were already evolving somewhat differently before the surge period. That is why the project avoids causal language and instead emphasizes measurement and KPI interpretation.

Pre-period group means

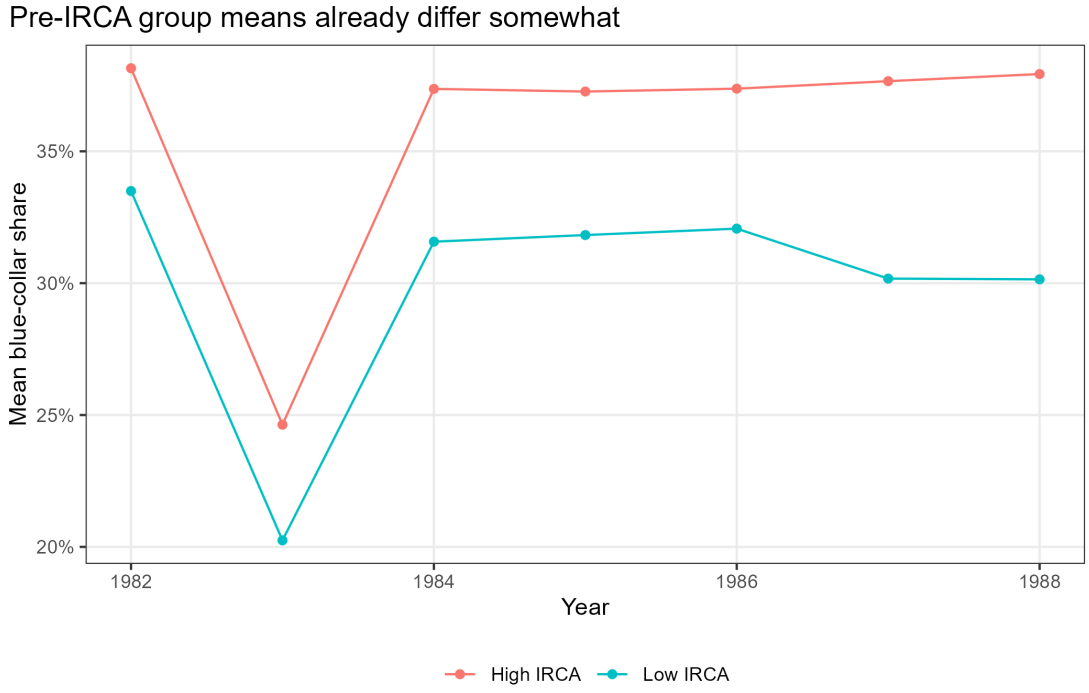

Figure 8: Pre-IRCA group means for the blue-collar share.

Supporting Diagnostics

Exposure distribution

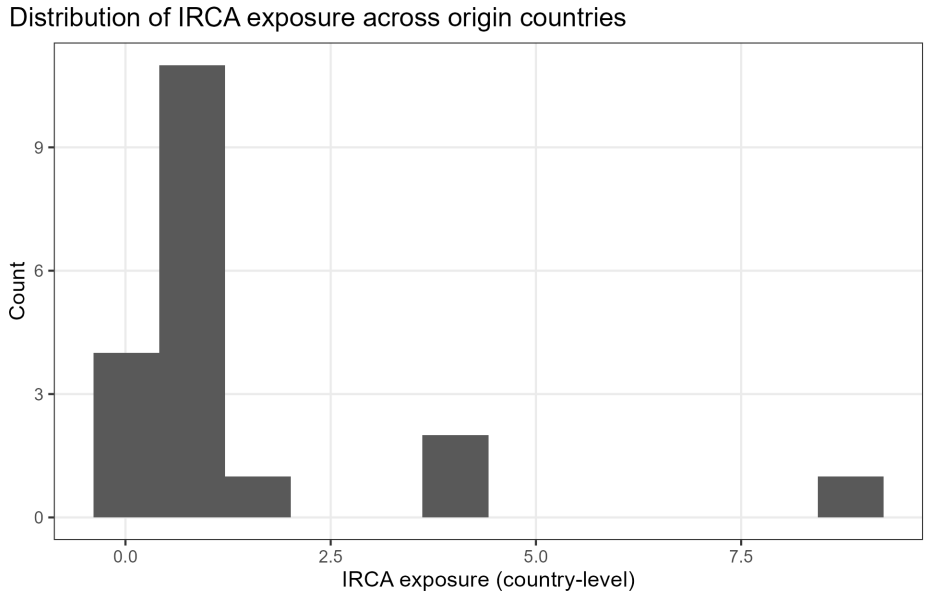

Figure 9: Distribution of IRCA exposure across origin countries.

Exposure versus change in blue share

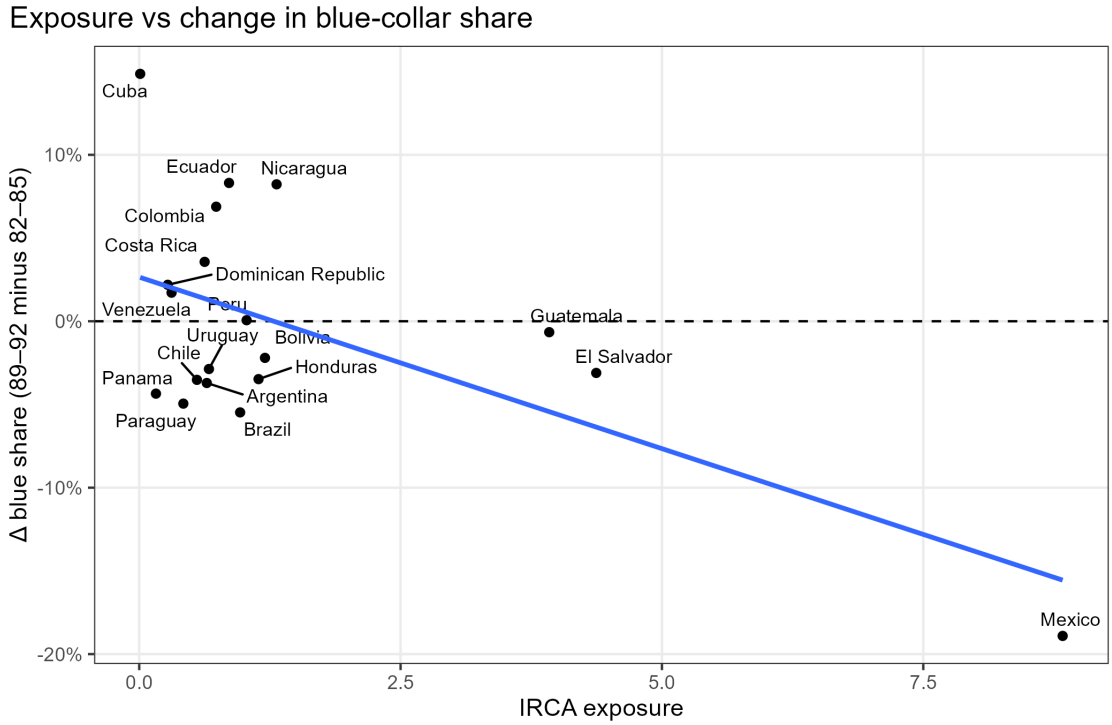

Figure 10: Country-level exposure versus change in the blue-collar share between pre-period and IRCA years.

These visuals help explain the structure of the sample and the variation behind the panel regressions.

Reporting Recommendation

The most practical output of the project is a KPI recommendation. For any stakeholder-facing dashboard or monitoring workflow, report these together:

1. Blue share (net denominator)
2. Blue share (full denominator)
3. No-occupation share
4. Denominator count / coverage context

That package makes it much harder to mistake a denominator artifact for a real compositional shift.

What I would do in production

If this KPI were deployed in a real reporting workflow, I would add simple rules such as:

- flag cases where the net share moves but the full share does not,

- flag large changes in the no-occupation share,
- require denominator counts to be reviewed before interpreting share movements,
- document the denominator definition directly in the dashboard.

Portfolio Takeaway

This project demonstrates the ability to:

- build a reproducible panel from messy administrative-style data,
- define and operationalize a country-level exposure measure,
- audit denominator sensitivity in a headline KPI,
- decompose a share into numerator and denominator channels,
- run robustness checks without overstating causal claims,
- translate technical output into a stakeholder-facing measurement recommendation.

That is what makes it a strong data analyst portfolio project.

How to Run

1. Put the raw Excel inputs in `data_raw/`
2. Open the project in R
3. Run `irca.R`
4. Render this `.Rmd`

Outputs are written to:

- `outputs/tables/`
- `outputs/figures/`
- `outputs/logs/`
- `data_clean/panel.rds`

Sources

- U.S. Immigration and Naturalization Service (INS). *Statistical Yearbook of the Immigration and Naturalization Service*, various years.

Appendix

Appendix A: Data quality

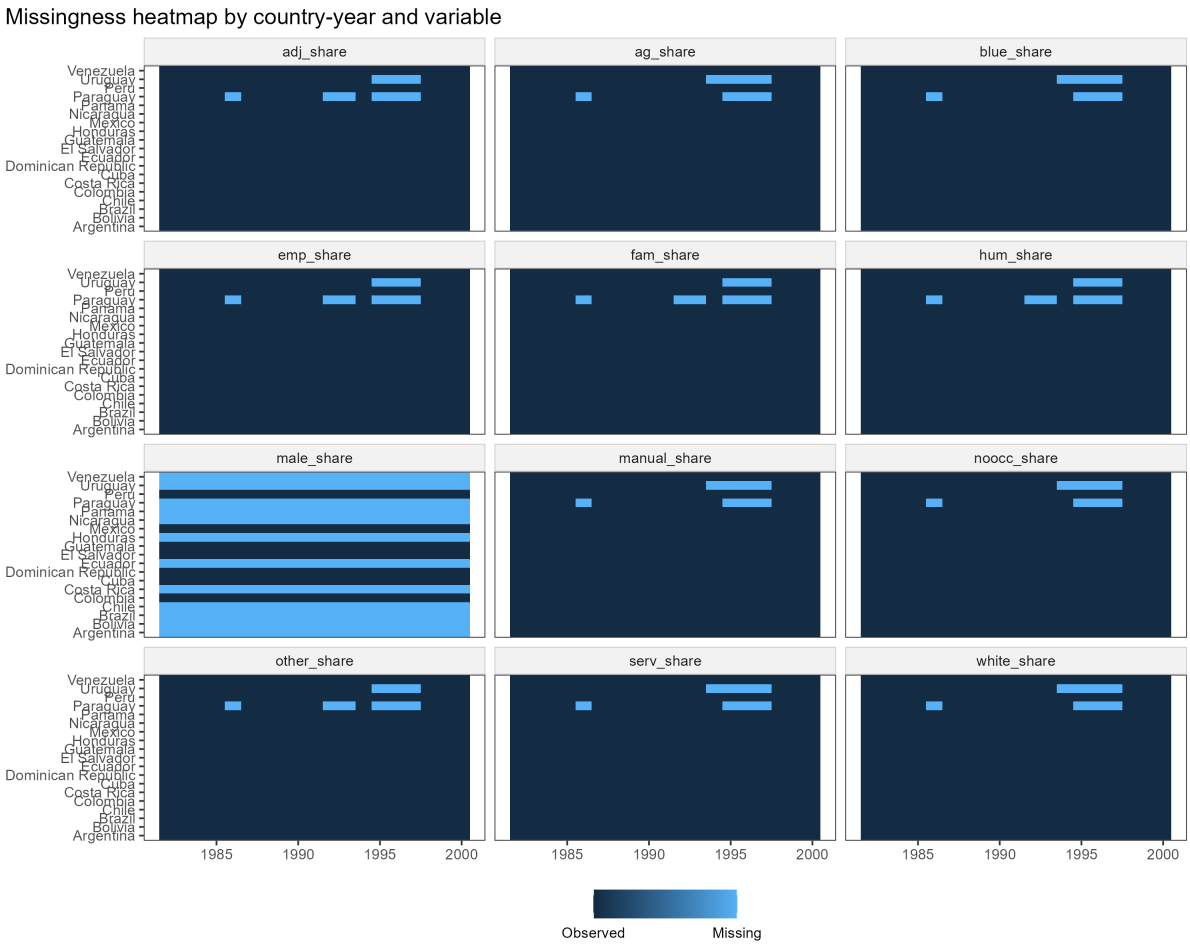

Figure 11: Missingness heatmap by country-year and variable.

Table 11: Missingness by variable at the country-year level.

Variable	Obs.	Missing	Total	Share missing (%)
Male share	133	228	361	63.2
Family share	352	9	361	2.5
Employment share	352	9	361	2.5
Humanitarian share	352	9	361	2.5
Other share	352	9	361	2.5
Adjustment share	352	9	361	2.5
Blue share	353	8	361	2.2
Agriculture share	353	8	361	2.2
Manual share	353	8	361	2.2
Services share	353	8	361	2.2
White share	353	8	361	2.2
No-occupation share	353	8	361	2.2

Table 12: Coverage by origin country for key variables.

Country	Years total	Years covered (key vars)	Coverage ratio
Argentina	19	19	1.000
Bolivia	19	19	1.000
Brazil	19	19	1.000
Chile	19	19	1.000
Colombia	19	19	1.000
Costa Rica	19	19	1.000
Cuba	19	19	1.000
Dominican Republic	19	19	1.000
Ecuador	19	19	1.000
El Salvador	19	19	1.000
Guatemala	19	19	1.000
Honduras	19	19	1.000
Mexico	19	19	1.000
Nicaragua	19	19	1.000
Panama	19	19	1.000
Peru	19	19	1.000
Venezuela	19	19	1.000
Uruguay	19	15	0.789
Paraguay	19	13	0.684

Appendix B: Baseline regression panels

Table 13: Baseline TWFE (Total): occupation shares.

	Blue	Agri	Manual	Services	White
IRCA intensity × Post	−0.025***	0.029***	0.004	−0.007**	0.004
	(0.005)	(0.004)	(0.005)	(0.003)	(0.005)
Num.Obs.	353	353	353	353	353
Mean of Y (share units)	0.342	0.050	0.392	0.224	0.384

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by origin-country. Shares in [0,1]; coefficients are pp changes per one-unit exposure. Total specs exclude adjustment-of-status share. Exposure stats (country-level): mean = 1.476, SD = 2.119.

Table 14: Baseline TWFE (Total): no-occupation share and class-of-admission shares.

	No-Occ	Family	Employment	Humanitarian	Other
IRCA intensity × Post	−0.035***	−0.052***	−0.021*	−0.004***	−0.002
	(0.007)	(0.004)	(0.011)	(0.001)	(0.001)
Num.Obs.	353	352	352	352	352
Mean of Y (share units)	0.620	0.542	0.301	0.067	0.013

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by origin-country. Shares in [0,1]; coefficients are pp changes per one-unit exposure. Total specs exclude adjustment-of-status share. Exposure stats (country-level): mean = 1.476, SD = 2.119.

Table 15: Baseline TWFE (Net): occupation shares controlling for adjustment share.

	Blue	Agri	Manual	Services	White
IRCA intensity × Post	−0.036***	0.029***	−0.007	−0.006*	0.013*
	(0.007)	(0.004)	(0.006)	(0.003)	(0.006)
Adjustment share	0.169*	−0.006	0.164*	−0.019	−0.144
	(0.085)	(0.027)	(0.082)	(0.032)	(0.093)
Num.Obs.	351	351	351	351	351
Mean of Y (share units)	0.342	0.050	0.392	0.224	0.384

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by origin-country. Shares in [0,1]; coefficients are pp per one-unit exposure. Net specs include adjustment-of-status share. Exposure stats (country-level): mean = 1.476, SD = 2.119.

Table 16: Baseline TWFE (Net): no-occupation share and class-of-admission shares controlling for adjustment share.

	No-Occ	Family	Employment	Humanitarian	Other
IRCA intensity × Post	−0.025***	−0.027**	−0.008	−0.016**	−0.009
	(0.005)	(0.011)	(0.011)	(0.006)	(0.009)
Adjustment share	−0.149*	−0.400**	−0.206	0.192**	0.108
	(0.081)	(0.152)	(0.128)	(0.091)	(0.138)
Num.Obs.	351	352	352	352	352
Mean of Y (share units)	0.619	0.542	0.301	0.067	0.013

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by origin-country. Shares in [0,1]; coefficients are pp per one-unit exposure. Net specs include adjustment-of-status share. Exposure stats (country-level): mean = 1.476, SD = 2.119.

Appendix C: Additional supporting outputs

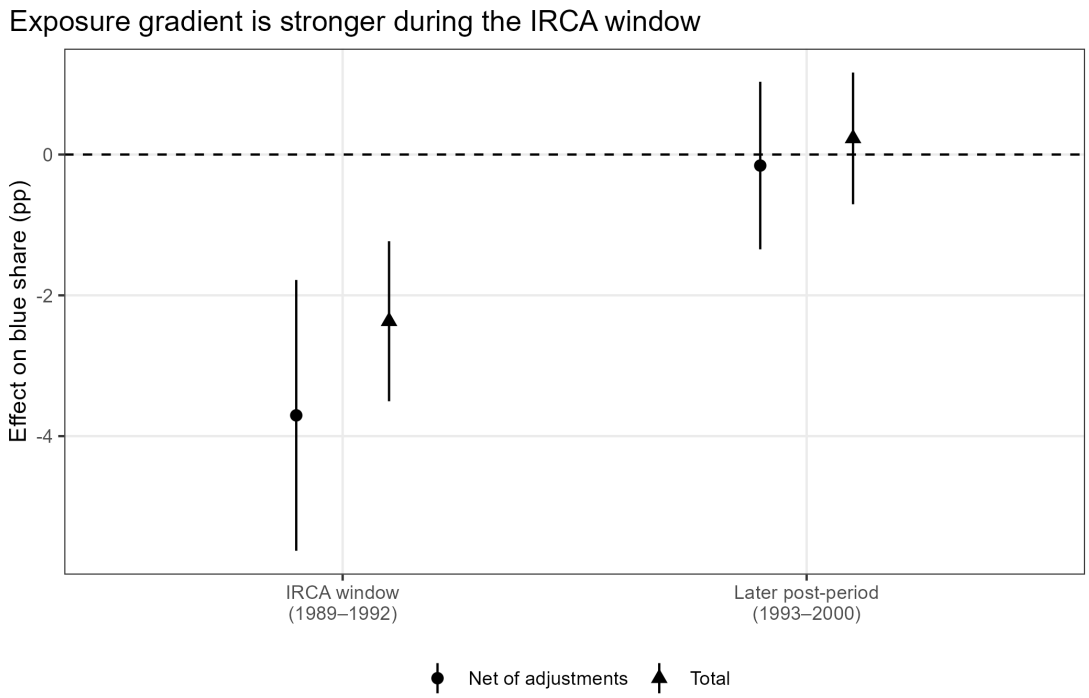

Figure 12: Early versus later post-period estimates for the blue-share result.

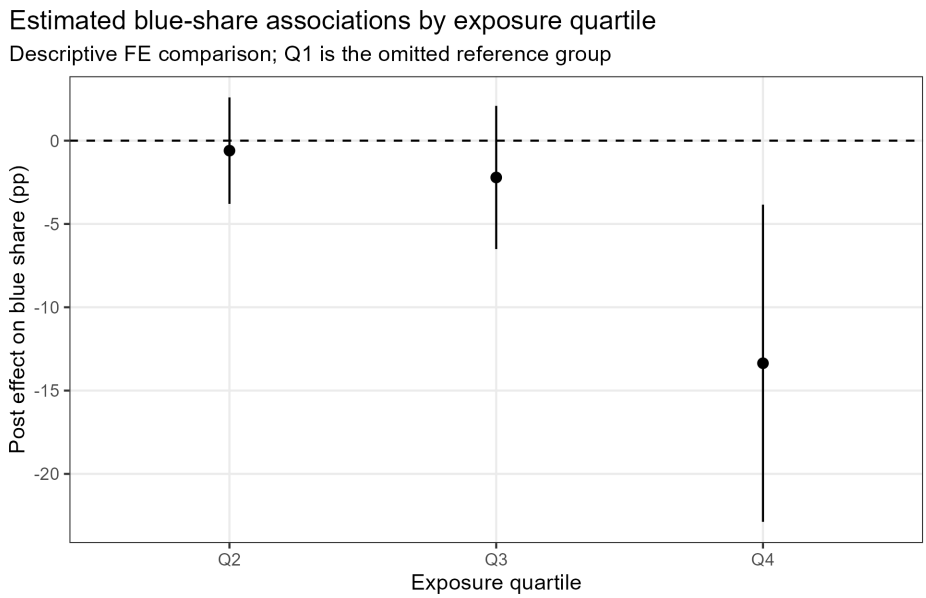

Figure 13: Estimated blue-share associations by exposure quartile.

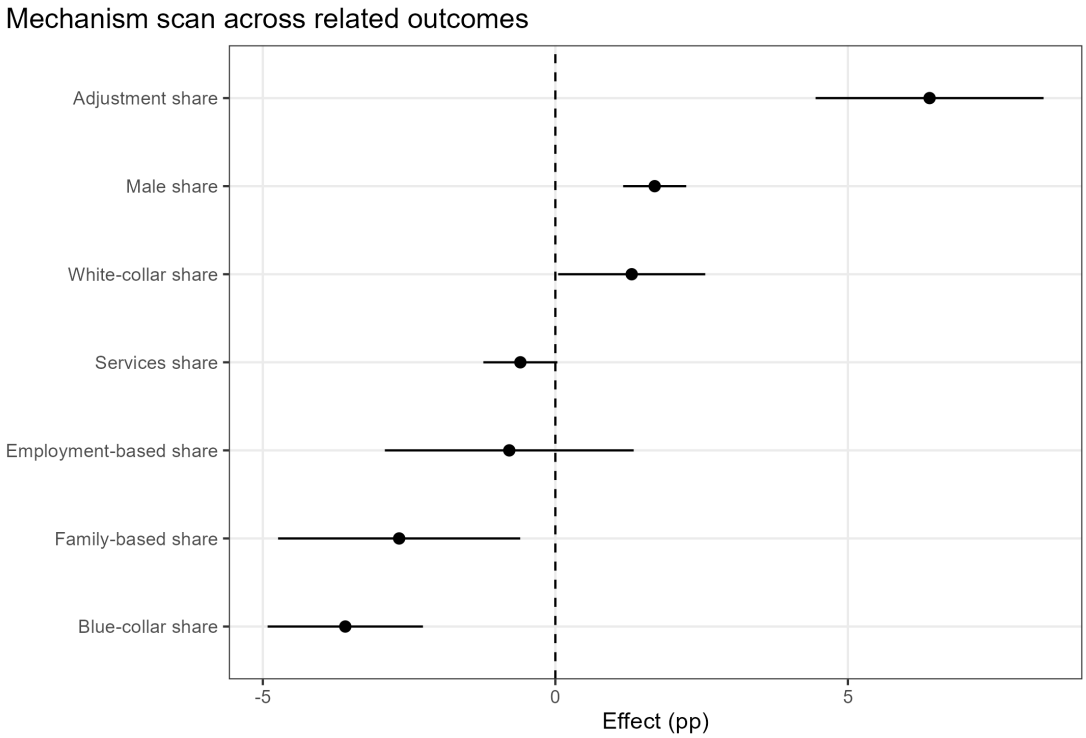

Figure 14: Mechanism scan across related outcomes.

Table 17: Alternative exposure construction and Mexico exclusion.

	Blue Net (baseline)	Blue Net (alt denom)	Blue Net (–MX)	Blue Net (alt –MX)
IRCA intensity × Post	−0.036***		−0.028**	
	(0.007)		(0.013)	
IRCA intensity (alt) × Post		−0.009***		−0.007**
		(0.002)		(0.003)
Adjustment share	0.169*	0.169*	0.163	0.163
	(0.085)	(0.085)	(0.106)	(0.106)
Num.Obs.	351	351	332	332

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by origin-country. Outcome is share in [0,1]; coefficients are pp.

Table 18: Weighted specifications.

	Net (unweighted)	Net (w: admissions)	Net (w: known occ.)
IRCA intensity × Post	−0.036***	−0.036***	−0.034***
	(0.007)	(0.003)	(0.004)
Adjustment share	0.169*	0.150**	0.105
	(0.085)	(0.069)	(0.084)
Num.Obs.	351	351	351

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by country.

Table 19: Balanced-panel restriction.

	Blue Net (balanced)
IRCA × Post	−0.037***
	(0.007)
Adjustment share	0.190*
	(0.090)
Num.Obs.	323

* p <0.1, ** p <0.05, *** p <0.01
Balanced panel: keeps only countries observed in all years of the Net sample.

Table 20: Robustness to origin-specific linear trends.

	Baseline Total	Baseline Net	Trends Total	Trends Net
IRCA intensity × Post	−0.025***	−0.036***	−0.025***	−0.036***
	(0.005)	(0.007)	(0.005)	(0.008)
Adjustment share		0.169*		0.160
		(0.085)		(0.099)
Num.Obs.	353	351	353	351

* p <0.1, ** p <0.05, *** p <0.01
Country & Year FE; SE clustered by origin-country. Trends specs include origin-specific linear trends (country[t]).